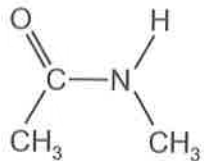


22 The structure of N-methylethanamide is shown.



N-methylethanamide

Which statements can be used to explain why an aqueous solution of N-methylethanamide has no effect on red litmus?

- 1 Delocalisation of the lone pair of electrons on the nitrogen atom over the C=O group results in the nitrogen becoming less available for protonation.
- 2 Delocalisation of the lone pair of electrons on the nitrogen atom over the C=O group strengthens the C–N bond.
- 3 N-methylethanamide is more stable than the ion that forms when it is protonated.

A 1 and 3

B 1 only

C 2 and 3

D 2 only